BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022 • Paper Code: 9221

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** 9221

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. The branch of biology which deals with the study of classification of organisms and their scientific names is called .....

(A) morphology
(B) anatomy
(C) taxonomy
(D) histology

Q2. Which bio-element makes most of the composition of organism's body?

(A) hydrogen
(B) carbon
(C) oxygen
(D) nitrogen

Q3. "If" and "then" logics are involved in .....

(A) observations
(B) experiments
(C) deductions
(D) conclusions

Q4. Rhizopus (bread mold) is an example of .....

(A) algae
(B) fungi
(C) plant
(D) bacteria

Q5. The zoological name of dog is .....

(A) felis domestica
(B) felis leo
(C) canis familiaris
(D) canis lupus

Q6. The number of chromosomes in human cell is .....

(A) 16 (B) 18

(C) 8 (D) 46

Q7. The tissue responsible for the transport of water and dissolved substances from roots to the upper parts is called ..... tissue.

(A) phloem (B) xylem

(C) epidermal (D) ground

Q8. Meiosis takes place in which of the following organs?

(A) testes (B) lungs

(C) heart (D) stomach

Q9. Photosynthesis is an example of .....

(A) catabolism (B) anabolism

(C) ingestion (D) digestion

Q10. Which one of the following pair of light colours is important for photosynthesis?

(A) green and yellow (B) yellow and orange

(C) green and orange (D) red and blue

Q11. Marasmus is a type of .....

(A) MDD (B) PEM

(C) OIN (D) SIM

Q12. Plasma protein that defends the body against pathogens is .....

(A) fibrinogen (B) albumins

(C) antibodies (D) pepsin

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32 | **Paper Code:** 9221

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) What level of organization is represented by volvox?

(ii) Describe the use of mathematical rules in biology.

(iii) What is classification? Briefly describe the principles of classifications.

(iv) Write down the functions of light microscope.

(v) Briefly explain that meiosis is essential for sexual reproduction.

(vi) Briefly describe how enzymes lower the activation energy.

(vii) What is the role of pigments during photosynthesis?

(viii) Discuss the role of dietary fiber in balanced diet.

(ix) Write down the characteristics of colon.

(x) Define pulse. Why is it important for human heart?

(xi) Briefly describe atherosclerosis (narrowing of arteries).

SECTION-C

Maximum Marks: 21

> Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) Describe career of biology in the field of agriculture and animal science.

(ii) Highlight the major reasons of deforestation.

Q3.

(i) Describe the structure of cell membrane with the help of fluid mosaic model.

(ii) Explain the significance of meiosis.

Q4.

(i) Describe the induced fit model of enzyme action.

(ii) Explain aerobic respiration.

Q5.

(i) Explain the role of oral cavity and pharynx in the human digestive system.

(ii) What is transpiration pull? Describe the importance of transpiration in the life of plants.